

Simple Harmonic Motion Calculations: A Beginner's Guide

Lesson 2: Period and Frequency: The Basics

Period and Frequency: The Basics - Study Notes

Key Concepts

- **Period (T):** The time taken for one complete oscillation or cycle. Measured in seconds (s).
- **Frequency (f):** The number of oscillations or cycles per unit time. Measured in Hertz (Hz).
- **Inverse Relationship:** Period and frequency are inversely proportional; as one increases, the other decreases.
- **Mass-Spring System:** The period of oscillation for a mass attached to a spring.
- **Simple Pendulum:** The period of oscillation for a simple pendulum.

Summary

Period and frequency are fundamental concepts in describing oscillating systems, particularly those exhibiting Simple Harmonic Motion (SHM). The period (T) represents the duration of a single complete oscillation – think of it as one full swing of a pendulum or one complete up-and-down motion of a mass on a spring. Frequency (f), on the other hand, tells us *how many* of these oscillations occur within a given time frame. A higher frequency means more oscillations per second.

These two quantities are intimately linked. They are inversely related, meaning that if the period increases, the frequency decreases, and vice versa. The mathematical relationship expressing this is $T = 1/f$ and $f = 1/T$. Different systems, like mass-spring systems and simple pendulums, have specific formulas to calculate their periods, which depend on their physical properties (mass, spring constant, length, and gravity).

Understanding period and frequency is crucial for analyzing and predicting the behavior of oscillating systems in various fields, from physics and engineering to music and everyday phenomena.

Important Points

- **Units:**
- * Period (T): seconds (s)
- * Frequency (f): Hertz (Hz) where $1 \text{ Hz} = 1 \text{ oscillation/second}$
- **Inverse Relationship:** $T = 1/f$ and $f = 1/T$
- **Mass-Spring System Period:** $T = 2\pi\sqrt{m/k}$
- * 'm' = mass (kg)
- * 'k' = spring constant (N/m)
- **Simple Pendulum Period:** $T = 2\pi\sqrt{L/g}$
- * 'L' = length of the pendulum (m)
- * 'g' = acceleration due to gravity (approximately 9.8 m/s^2)
- **Small Angle Approximation:** The pendulum formula ($T = 2\pi\sqrt{L/g}$) is only valid for small angles of displacement (typically less than 15 degrees).

- **Oscillation:** A repetitive back-and-forth motion.

Examples

Example 1: Pendulum Frequency

A pendulum completes 5 oscillations in 10 seconds. What is its frequency?

- **Solution:**
 - * First, calculate the period (T): $T = \text{time} / \text{number of oscillations} = 10 \text{ s} / 5 = 2 \text{ s}$
 - * Then, calculate the frequency (f): $f = 1/T = 1/2 \text{ s} = 0.5 \text{ Hz}$
 - * Therefore, the frequency of the pendulum is 0.5 Hz.

Example 2: Mass-Spring Period

A 0.5 kg mass is attached to a spring with a spring constant of 200 N/m. What is the period of oscillation?

- **Solution:**
 - * Use the formula: $T = 2\pi \sqrt{m/k}$
 - * Substitute the values: $T = 2\pi \sqrt{0.5 \text{ kg} / 200 \text{ N/m}}$
 - * Calculate: $T = 2\pi \sqrt{0.0025} = 0.099 \text{ s} \approx 0.1 \text{ s}$ (approximately)
 - * Therefore, the period of oscillation is approximately 0.1 seconds.

Quick Review

1. What is the difference between period and frequency?
2. What are the units for period and frequency, respectively?
3. If the frequency of an oscillating system is 2 Hz, what is its period?
4. How does increasing the length of a simple pendulum affect its period (assuming small angles)?
5. Write down the formula for the period of a mass-spring system.

Further Reading

- **Simple Harmonic Motion (SHM):** Explore the characteristics and equations of SHM in more detail.
- **Damped Oscillations:** Investigate what happens when energy is lost from an oscillating system.
- **Driven Oscillations:** Learn about systems that are forced to oscillate at a specific frequency.
- **Wave Phenomena:** Understand how period and frequency relate to wave properties like wavelength and speed.
- **Resonance:** Explore the phenomenon of resonance and its applications.

